

11	THERMAL INSULATION			Technical Compliance / Deviations to be filled
11.1	Supply, Installation of Insulation material. Thermal insulation of Ducts with low smoke/low fire propagating FM listed closed cell nitrile rubber insulation having ASTM E84 (25/50) ratings with Class O and UL94(V0 & HB) approvals. Insulation to be factory laminated with minimum 12 microns thick glass reinforced pure aluminium foil of 3x3 mesh pattern with min k value of 0.036 w/mK @ 24 Deg C with working temp range till 85 deg C and have μ value of 12,000 as per EN12086. For outdoors, insulation to be factory laminated with laminated with Grey colored 400 microns/380 gsm multiple layered laminate of polymeric material reinforced with scrim with μ value of 20,000 and UV resistance as per ISO 4892-2 Method A (1000 Hrs test) and mechanical strength >5 MPa as per ASTM. Application to be done in square box method. Good bonding Min 75 mm width alu foil to be applied at each sides. To avoid condensation, Minimum 100 mm distance to be provided between insulated surfaces/walls for ventilation. For good workmanship and lower wastage, the Installation team to be certified by Manufacturer team for proper application with Min 2 day installation training to be conducted by manufacturer's person only. Adhesive to be used should be low voc and IGBC compliant and specially formulated for ducting insulation by OEM (SR 505/Duct Seal). Also, application to be done in well ventilated area with RH% below 80%. A periodic monthly site visit to be done by the manufacturer representative to ensure proper application quality. Report to be submitted to the user, pmc and consultant. Above mentioned specifications to be backed up by Third party test certificates from foreign/nabl accredited labs and to be submitted to PMC/Consultant for the approval before material supply. A random lab test to be conducted by user for the supplied product for K value, covering thickness, and ASTM E84(25/50) certifications at foreign/nabl accredited labs. Cost for the same to be borne by the contracting agency.			Comply with Class O
11.1.1	50 mm thick for Clean Room SA Duct - MAU Plenum	Sqm.	2500	
11.1.2	32 mm thick for Clean Room SA Duct - For 15 Meter from MAU	Sqm.	9000	
11.1.3	25 mm thick for Clean Room SA Duct - For After 15 Meter All Ducts	Sqm.	42000	
11.1.4	19 mm thick for Comfort SA Duct	Sqm.	9500	
11.1.5	13 mm thick for Comfort RA Duct	Sqm.	9500	
11.2	Supply, Installation of Insulation material. Acoustic Insulation of Duct Internal linings starting from AHU Collars till 5-7 meters by Dust and Fiber free Open Cell Nitrile Rubber with EPA approved antimicrobial additive for antimicrobial/anti viral properties as per EN ISO 846/JIS Z 280, having min 140-180 kg/m³ density and fire flammability as per UL94 HB and V0 along with min NRC 0.55 as per ASTM C423. For material integrity, the Insulation shall pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071 - 05. Adhesive to be used should be low voc and IGBC compliant and specially formulated for ducting insulation by OEM(SR 505 or Equivalent). Insulation to be approved by local fire authorities. Also, application to be done in well ventilated area with RH% below 80% For good workmanship and lower wastage, the Installation team to be certified by Manufacturer team for proper application with Min 2 day installation training to be conducted by manufacturer's person only. A periodic monthly site visit to be done by the manufacturer representative to ensure proper application quality. Report to be submitted to the PMC and User. Above mentioned specifications to be backed up by Third party test certificates from foreign/nabl accredited labs and to be submitted to PMC/Consultant for the approval before material supply. A random lab test to be conducted by user for the supplied product for density, and antimicrobial behaviour at foreign/nabl accredited labs. Cost for the same to be borne by the contracting agency. Thickness - Min 25 mm	Sqm.	2500	Comply with Class O
6	Supply & Installation of underdeck insulation as per specification. Material shall be 19mm thick Class " 0 " Closed cell nitrile rubber elastomeric foam insulation laminated with chemically treated glass cloth of 7 mil / 0.18 mm thickness Thermal conductivity of insulation material shall not exceed 0.036 W/(m.K) at mean temperature of 24C as per ASTM C 518 / ASTM C 177. The insulation material shall have water absorption of 0.2% by volume as per ASTM C 1763 ,shall have fire performance rating of 25/50 as per ASTM E 84. Density will be 40 to 60 Kg/m³. • The insulation material shall have Microban®* an anti-microbial product, which is EPA (Environmental Protection Agency), USA Approved, as an integral part of insulation that cannot be washed off or worn off.The insulation material shall give enhanced level of protection against harmful Microbes such as fungi and mold and should confirm to ASTM G 21, UL 181 & ASTM E 2180. Water Vapor Permeability of insulation material shall be minimum 0.05 Perm-inch. without any external barrier as per ASTM E 96 Procedure A. The adhesives for the application of insulation should be of principal manufacturer. Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation	Sqm	300	Comply with Class O
8	S.I.T.C. of Thermal insulation of CHW Piping with low fire propagation Class O Closed Cell Nitrile Rubber with K value of 0.035 w/m.k at 0 Deg C Temp, and UL94 certified from UL lab. Insulation to be laminated with Grey colored 400 microns/380 gsm multiple layered laminate of polymeric material reinforced with scrim with μ value of 20,000 and UV resistance as per ISO 4892-2 Method A (1000 Hrs test) and mechanical strength >5 MPa as per ASTM. To avoid condensation, Minimum 100 mm distance to be provided between insulated surfaces/walls for ventilation.For good workmanship and lower wastage, the Installation team to be certified by Manufacturer team for proper application with Min 2 day installation training to be conducted by manufacturer's person only. Adhesive to be used should be low voc and IGBC compliant and specially formulated for ducting insulation by OEM(SR 998 or Equivalent). Also, application to be done in well ventilated area with RH% below 80%. A periodic monthly site visit to be done by the manufacturer representative to ensure proper application quality. Report to be submitted to the PMC, Consultant and User / Internal Chiller Plant room Insulation shall be with 24G Aluminium Cladding Above mentioned specifications to be backed up by Third party test certificates from foreign/nabl accredited labs and to be submitted to PMC/Consultant for the approval before material supply. A random lab test to be conducted by user for the supplied product for K value, covering thickness/gsm, μ value and Class O certifications at foreign/nabl accredited labs. Cost for the same to be borne by the contracting agency. Price shall include all the accessories and PUF Block support of suitable density as per pipe and fixing arrangement with Structural support			Comply
8.1	50 mm thick Insulation	Sq. Mtr.	7,500.00	

8.2	44mm thick Insulation	Sq. Mtr.	8,500.00	
8.3	38 mm thick Insulation	Sq. Mtr.	4,000.00	
8.4	32 mm thick Insulation	Sq. Mtr.	7,000.00	
8.5	25mm Thick Insulation	Sq. Mtr.	4,500.00	
8.6	19mm Thick Insulation	Sq. Mtr.	1,500.00	
8.7	13mm Thick Insulation	Sq. Mtr.	150.00	